

HW02 逻辑代数基础OK!

1、一个三变量**非一致**判断电路，当输入的 3 个变量 A、B、C 不完全相同时，输出 $F=1$ ，否则 $F=0$ 。试列出该逻辑问题的真值表，并写出函数表达式。

参考解答：真值表如下：

A	B	C	F	A	B	C	F
0	0	0	0	1	0	0	1
0	0	1	1	1	0	1	1
0	1	0	1	1	1	0	1
0	1	1	1	1	1	1	0

$$F = (A + B + C)(\bar{A} + \bar{B} + \bar{C})$$

$$\text{Or } F = \bar{A}\bar{B}C + \bar{A}B\bar{C} + \bar{A}BC + A\bar{B}\bar{C} + A\bar{B}C + ABC\bar{C}$$

2、直接写出下列各函数的反函数和对偶式：

$$(1) F = AB + \bar{C}D + \overline{BC + \bar{D} + \bar{C}E + \bar{D} + E}$$

参考解答： $F' = (A + B)(\bar{C} + D)(B + C)\bar{D}(\bar{C} + E)\overline{\bar{D}E}$

$$\bar{F} = (\bar{A} + \bar{B})(C + \bar{D})(\bar{B} + \bar{C})D(C + \bar{E})\overline{\bar{D}E}$$

$$\bar{F} = (\bar{A} + \bar{B})(C + \bar{D})(BC + \bar{D} + \bar{C}E + \overline{\bar{D} + E})$$

$$F = AB + \bar{C}D + \overline{BC + \bar{D} + \bar{C}E + \bar{D} + E}$$

$$= AB + \bar{C}D + \overline{BC + \bar{D} + \bar{C}E + \bar{D}\bar{E}}$$

$$= AB + \bar{C}D + \overline{BC + \bar{D} + \bar{C}E}$$

注: $= AB + \bar{C}D + (\bar{B} + \bar{C})D(C + \bar{E})$

$$= AB + \bar{C}D + \bar{B}CD + \bar{C}D\bar{E}$$

$$= AB + \bar{C}D + \bar{B}CD$$

$$= AB + \bar{C}D + \bar{B}D$$

$$F' = (A + B)(\bar{C} + D)(\bar{B} + D)$$

$$\bar{F} = (\bar{A} + \bar{B})(C + \bar{D})(B + \bar{D})$$

$$(2) F = \overline{A + C(\bar{B}C + D)(B + C)} + AD$$

参考解答: $F' = (\overline{AC + B + \bar{C}D + BC})(A + D)$

$$\bar{F} = (\overline{\bar{A}\bar{C} + \bar{B} + \bar{C}\bar{D} + \bar{B}\bar{C}})(\bar{A} + \bar{D})$$

$$\bar{F} = (\overline{A + C(\bar{B}C + D)} + \bar{B} + \bar{C}) \cdot \bar{A}\bar{D}$$

$$F = \overline{\overline{A+C}(\overline{BC+D})(B+C) + AD}$$

注：

$$= ((A+C) + BCD)(B+C) + AD$$

$$= (A+C)(B+C) + AD$$

$$= C + AB + AD$$

$$(3) F = A \cdot \overline{B + \overline{D}} + (AC + BD)E$$

参考解答： $F' = (A + \overline{B\overline{D}})((A+C)(B+D) + E)$

$$\overline{F} = (\overline{A} + \overline{\overline{B\overline{D}}})((\overline{A} + \overline{C})(\overline{B} + \overline{D}) + \overline{E})$$

$$\overline{F} = (\overline{A} + B + \overline{D})(\overline{AC + BD + E})$$

$$F = A \cdot \overline{B + \overline{D}} + (AC + BD)E$$

注：

$$= A\overline{B}D + ACE + BDE$$

$$= A\overline{B}D + ACE + BDE$$

3、写出逻辑函数 F 的标准与-或式：

$$(1) F = \overline{B}\overline{C} + BC + A$$

参考解答：

$$F = \overline{B}\overline{C} + BC + A$$

$$= (\overline{A} + A)\overline{B}\overline{C} + (\overline{A} + A)BC + A(\overline{B} + B)(\overline{C} + C)$$

$$= \overline{A}\overline{B}\overline{C} + A\overline{B}\overline{C} + \overline{A}BC + ABC + (A\overline{B} + AB)(\overline{C} + C)$$

$$= \overline{A}\overline{B}\overline{C} + A\overline{B}\overline{C} + \overline{A}BC + ABC + (A\overline{B} + AB)\overline{C} + (A\overline{B} + AB)C$$

$$\begin{aligned}
&= \bar{A}\bar{B}\bar{C} + A\bar{B}\bar{C} + \bar{A}BC + ABC + A\bar{B}C + AB\bar{C} + A\bar{B}C + ABC \\
&= \bar{A}\bar{B}\bar{C} + A\bar{B}\bar{C} + \bar{A}BC + ABC + AB\bar{C} + A\bar{B}C \\
&= \sum m(0, 3, 4, 5, 6, 7)
\end{aligned}$$

$$(2) F = BD + ACD + AB\bar{D} + A\bar{C}\bar{D}$$

参考解答： $F = BD + ACD + AB\bar{D} + A\bar{C}\bar{D}$

$$\begin{aligned}
&= (\bar{A} + A)B(\bar{C} + C)D + A(\bar{B} + B)CD + AB(\bar{C} + C)\bar{D} + A(\bar{B} + B)\bar{C}\bar{D} \\
&= \bar{A}\bar{B}\bar{C}D + A\bar{B}\bar{C}D + \bar{A}BCD + ABCD + A\bar{B}CD \\
&\quad + ABCD + AB\bar{C}\bar{D} + ABC\bar{D} + A\bar{B}\bar{C}\bar{D} + AB\bar{C}\bar{D} \\
&= \bar{A}\bar{B}\bar{C}D + A\bar{B}\bar{C}D + \bar{A}BCD + ABCD + A\bar{B}CD \\
&\quad + AB\bar{C}\bar{D} + ABC\bar{D} + A\bar{B}\bar{C}\bar{D}
\end{aligned}$$

$$= \sum m(5, 7, 8, 11, 12, 13, 14, 15)$$

$$(3) F = \overline{AB} + A\overline{B}(AC + CD)$$

参考解答: $F = \overline{AB} + A\overline{B}(AC + CD)$

$$= \overline{A \oplus B}(AC + CD)$$

$$= (A \odot B)(AC + CD) = (AB + \overline{A}\overline{B})(AC + CD)$$

$$= ABAC + ABCD + \overline{A}\overline{B}AC + \overline{A}\overline{B}CD$$

$$= ABC + ABCD + \overline{A}\overline{B}CD$$

$$= ABC\overline{D} + ABCD + \overline{A}\overline{B}CD$$

$$= \sum m(3, 14, 15)$$

$$(4) F = A \oplus \overline{B \oplus C}$$

参考解答: $F = A \oplus \overline{B \oplus C}$

$$= \overline{A} \overline{B \oplus C} + A(B \oplus C)$$

$$= \overline{A}(B \odot C) + A(B \oplus C)$$

$$= \overline{A}(BC + \overline{B}\overline{C}) + A(\overline{B}C + B\overline{C})$$

$$= \overline{A}BC + \overline{A}\overline{B}\overline{C} + A\overline{B}C + AB\overline{C}$$

$$= \sum m(0, 3, 5, 6)$$

4、用真值表证明:

$$(1) AB + \bar{A}C + BC = AB + \bar{A}C$$

参考解答:

A	B	C	左边	右边	A	B	C	左边	右边
0	0	0	0	0	1	0	0	0	0
0	0	1	1	1	1	0	1	0	0
0	1	0	0	0	1	1	0	1	1
0	1	1	1	1	1	1	1	1	1

$$(2) \bar{A}\bar{B}C + \bar{A}B\bar{C} + A\bar{B}\bar{C} + ABC = A \oplus B \oplus C$$

参考解答:

A	B	C	左边	右边	A	B	C	左边	右边
0	0	0	0	0	1	0	0	1	1
0	0	1	1	1	1	0	1	0	0
0	1	0	1	1	1	1	0	0	0
0	1	1	0	0	1	1	1	1	1

5、用代数法证明：

$$(1) A \oplus B \oplus C = A \odot B \odot C$$

参考解答： $A \oplus B \oplus C$

$$= \bar{A}(B \oplus C) + A \cdot \overline{B \oplus C}$$

$$= \bar{A} \cdot \overline{B \odot C} + A(B \odot C)$$

$$= A \odot B \odot C$$

$$(2) \bar{B}\bar{D} + BD + AB = \bar{B}\bar{D} + BD + A\bar{D}$$

参考解答: $\bar{B}\bar{D} + BD + AB$

$$= A\bar{B}\bar{D} + \bar{B}\bar{D} + BD + ABD + AB\bar{D}$$

$$= A\bar{B}\bar{D} + \bar{B}\bar{D} + BD + AB\bar{D}$$

$$= \bar{B}\bar{D} + BD + A\bar{D}$$

$$(3) \overline{A + B + \bar{C}} \cdot \bar{C}D + (B + \bar{C})(A\bar{B}D + \bar{B}\bar{C}) = 1$$

参考解答：

$$\begin{aligned}
 & \overline{\overline{A+B+\bar{C}} \cdot \bar{C}D} + (B+\bar{C})(A\bar{B}D + \bar{B}\bar{C}) \\
 &= (A+B+\bar{C}) + (C+\bar{D}) + (B+\bar{C})(A\bar{B}D + \bar{B}\bar{C}) \\
 &= A+B+\bar{C} + C+\bar{D} + (B+\bar{C})(A\bar{B}D + \bar{B}\bar{C}) \\
 &= A+B+1+\bar{D} + (B+\bar{C})(A\bar{B}D + \bar{B}\bar{C}) = 1
 \end{aligned}$$

(4) $(B+C)(C+D)$

$$= (A+B+C)(B+C+\bar{D})(C+D+\bar{E})(C+D+E)(A+B+C+D+F)$$

参考解答：

$$\begin{aligned}
 & (A+B+C)(B+C+\bar{D})(C+D+\bar{E})(C+D+E)(A+B+C+D+F) \\
 &= (A+B+C)(B+C+\bar{D})(C+D)
 \end{aligned}$$

$$\begin{aligned}
 &= (B + C + A\bar{D})(C + D) \\
 &= C + D(B + A\bar{D}) \\
 &= C + BD = (B + C)(C + D)
 \end{aligned}$$

6、已知 $F(A, B, C, D) = \prod M(0, 2, 3, 4, 7, 8, 10)$ ，写出 F 的标准与或式以及 $\bar{F}$ 、 F' 的所有标准式。

参考解答： $F(A, B, C, D) = \sum m(1, 5, 6, 9, 11, 12, 13, 14, 15)$

$$\bar{F}(A, B, C, D) = \sum m(0, 2, 3, 4, 7, 8, 10)$$

$$\bar{F}(A, B, C, D) = \prod M(1, 5, 6, 9, 11, 12, 13, 14, 15)$$

$$F'(A, B, C, D) = \sum m(5, 7, 8, 11, 12, 13, 15)$$

$$F'(A, B, C, D) = \prod M(0-4, 6, 9, 10, 14)$$